

ANSWER TO REVIEWERS.

I would like to address a sincere word of thanks to the anonymous reviewers for their constructive and insightful comments to provide valuable feedback on our article. I paid heed to the questions, pieces of advice and suggestions made by the reviewers and have accordingly modified the manuscript.

To ease the reading of the responses, I have adopted the following color convention: in black the review and in blue my response.

Reviewer: 1

General remarks

In this paper, the author introduces BooN, an open-source Python-based software designed for Boolean network modelling. The key contribution is the comprehensive integration of tools for Boolean network design, dynamics modelling, symbolic computation of stable states, and controllability analysis. In addition, all these functions are available via a graphical user interface, supported by a tutorial on how to use it. In this way, the method provides a facility offering both user-friendly and technically robust functionality, addressing the needs of both beginners and advanced users in biological network modelling.

The method's key contribution is its controllability analysis, which is crucial for studying gene regulatory networks in pathologies or drug target discovery. Moreover, its integration into the Colomoto framework enhances its usability and interoperability with other network analysis tools. Finally, the tool supports multiple input formats, making it adaptable for researchers working with different data formats.

I will list below some points for improvement to further strengthen its impact:

While the author mentions the availability of documentation and examples, more detail on real-world case studies or an extended example (beyond breast cancer) could enhance understanding of how BooN performs in practice.

We have completed the jupyter code [breast-cancer-study.ipynb](#) related to breast cancer study, by adding comments concerning the explanation of the analysis, and also for the example. I hope this complement will be sufficient.

The article could be more impactful if performance comparison is provided, especially on large-scale networks. While the author highlights the tool's limitations for analysing larger cyclic

attractors, explicit details on computational efficiency compared to other tools would give users a clearer sense of BooN's scalability.

I have added a section detailing the benchmarks of the main method of BooN.

The software currently supports Boolean control networks with node-level interventions. Future work could explore extending control options to more refined network components, such as edges, which would add flexibility for modelling complex biological interactions.

Effectively the method is restricted to nodes. However, we had already developed the computational framework to control edges. We do not include it because the control on edge does not seem to be practically used for biological application. For example, in drug discovery the issue is to identify a gene, and not its interaction. However, thanks to your comment we plan to extend the controllability on edges in the future, since you think it can be useful.

Concerning the GitHub repository:

The jupyter notebooks `breast-cancer-study.ipynb` and `example.ipynb` need more explanations on each step of the analysis. In addition, in these examples, building the interaction graph is done near the end of the analysis; I suggest this part be moved just after importing or introducing the regulatory network.

We have completed the jupyter code `breast-cancer-study.ipynb` related to breast cancer study, by adding comments concerning the explanation of the analysis, and also for the example. We also move the interaction graph in the network analysis at the beginning of the program. .

Grammar suggestions/corrections:

We have applied the correction. Thank you for your suggestions.

Adding the hyperlink to the GitHub repository would facilitate the tool's accessibility.

I have added a link to access to the site directly.

Reviewer: 2

Comments to the Author

[Review] BooN: Boolean Network Analysis Software

The application note presents the BooN toolbox, which supports manipulation of Boolean networks, symbolic computation of fixed-points, explicit computation of complex attractors, and identification of permanent control interventions. According to the author, the main novel contributions of BooN are (broadly speaking) its controllability method and its graphical user

interface. Overall, these are not exceptionally novel (e.g. other controllability/GUI tools exist), but I think BooN is a welcome addition into the ecosystem of Boolean network tooling and adds to the diversity of available methods and formats. However, I have some reservations regarding the overall presentation and quality of the tool that I think are easily addressable and would make this a much more interesting submission.

Comments regarding text

Overall, I think the text spends a lot of time explaining what Boolean networks are and how they work, but not that much is actually said about BooN itself. The BooN section + Conclusion + Fig. 4 are only one page of the whole text, the rest is introduction and theory. I understand that the work needs to be properly motivated and important concepts must be explained. But with a 5 page limit, I believe that at least some of the theory can be perhaps simplified or moved to a supplementary material to provide more room for talking about BooN.

Instead, what I would suggest:

- * There is no mention of performance or benchmarks anywhere in the text. Even if a proper benchmark is not available, at least some discussion of BooN's capabilities in that regard would be welcome.

We have added a section for the BooN benchmark evaluating the primary methods of BooN: stable state computation, brute force equilibria computation and controllability analysis.

- * Maybe some extra discussion can be dedicated to the control method. I understand that it is already published, but a more comprehensive summary would be interesting.

The control analysis is fully detailed in [Biane2019]. Essentially, we summarize the approach. In response to your comment, I have tried to add some elements explaining the controllability, notably its application in the article, but I do not really know what I can add which is not already published concerning the method. Moreover, I have added a sentence to refer to this article for the theoretical background study.

For example, you mention that the method identifies negative control parameters, but what about positive control parameters?

Concerning the negative control parameters, the control is active when its value $= 0$. This is a convention motivated by the fact that the control freezes the variable and 0° is the freeze temperature ;-).

Therefore, since we need to define which control parameters are active, (ie. equal to 0) we need to extract negative control parameters in the result of the prime implicant computation which is a minterm because these terms are true only when the control is false. Moreover, the positive control parameters thus correspond to inactive control, and thereby they are not considered since we focus on the active control.

I have given more explanation in the article to explain this element.

* Perhaps more could be also said about what is available in the Boon and logic Python modules? You mentioned that the Boolean network is completely defined by the interaction graph. How does this work for complex, high arity functions?

The conversion of the interaction graph with the specification of modules (integer identifier) leads to a DNF formula. There is no limitation of size and all types of formulas can be encoded since it was a DNF except the graphical limitation for drawing very large graph. We have completed the explanation in the article.

Are there some functions that you cannot represent in this way? Do you use some existing libraries to provide the SBML support?

BooN is able to read SBML file but not generate them (yet). We use the standard SBML package libsbml.

* Another option could be a small biological case study that illustrates the usefulness of controllability analysis and the fact that BooN can be applied to realistic biological models. Maybe even implementing the example network (i.e. equations (1) and (2)) in BooN and showing the result would be interesting.

The usefulness of the controllability analysis is fully illustrated in the example breast-cancer study.py. This example was also fully described in our article which is dedicated to controllability analysis previously mentioned.

A sentence was added to the article to direct readers interested in concrete applications of controllability to the example included in the distribution.

* (However, do note that these are only suggestions, I don't think it is necessary to add all of this)

* The references and related work could be better:

* As far as I can tell, application notes have a limit of 10 references, this text has 20. However, this limit is often exceeded in other application notes that I found published in this journal, so I don't consider this to be a major problem.

* With that said, references 5-12 (except 7) are only used to "generally" introduce the controllability problem, but are (as far as I can tell) not directly related to methods implemented by Boon, or not directly compared to Boon. I'm by no means saying these are irrelevant, but with a limit like this, I would expect some more explicit connection to the tool itself.

We have removed the citations you have mentioned mainly used to introduce the controllability but which are not discussed and compared. A full state of art comparison of the method and those mentioned here was achieved in the article [Biane2019].

* Meanwhile, BoolNet is not cited properly, even though it has an associated publication [Mussel2010], and other tools from the Colomoto docker that are similar in terms of capabilities are also not mentioned.

The reference is added..

For example, pyboolnet [Klarner2017] also supports Boolean network manipulation, computation of attractors, and prime implicants (it does not support control as far as I know). Also AEON.py [Benes2022] is another library for Boolean network manipulation, symbolic attractor/steady state computation and control (I believe it also has a web GUI?). Both are available in Python.

In accordance with your request, I have incorporated the two references that you specified. I am grateful to you for your comprehensive and interesting review. Concerning AON I do not see any reference to a Graphical interface in the documentation.

Other minor issues

* The first reference would really benefit from having a URL. I can find the repository based on its name, but it is somewhat tedious.

I have added an hyperlink

* The sentence on lines 8-10 on page 2 does not make much sense. Seems like some content is missing?

I have proposed an alternative sentence. I hope, this will be clearer. Otherwise I suggest to remove it since it is not central.

* S_X (line 22, page 2) is not defined anywhere, but considering what I said about simplifying the theory section, I would just replace it with $\mathbb{B}^n$?

We define S_X which is the set of all logical interpretations corresponding to functions assigning Boolean value to variables. Now, I have defined S_X before.

* Listing the exact queries in Fig. 1 also seem rather unnecessary for this kind of paper. Personally, I would move this to a supplement.

We remove them.

* Page 2, line 34: To me, "attractors reduced to a single state" implies that the attractors are somehow modified (by a reduction) to obtain one state. Maybe "corresponding to a single state" would be better

We rephrase the sentence according to your suggestion

* I believe it's written i.e., not ie.?

Actually it was a mistake in my \ie command, I correct it. thank you.

Comments regarding tool

* I was able to install the tool and run the example/case study without any issues. I especially appreciate the provided documentation. I was also able to load a somewhat larger network (285 variables) and compute its stable states.

* With that said... the readme and documentation don't say how to start the GUI? I eventually figured it out, but for new users, this is probably something you want to explain.

I have corrected this in the README

* The GUI works... but I have had some issues with usability.

* If I press enter repeatedly while a node or edge isn't selected (i.e. as if adding label to a node), the app prints an error into the command line.

Sometimes it recovers without issues, but it seems that depending on other actions I performed (or other app windows I have open?), this can break the option to edit update functions or to kill/freeze the app outright.

Thank you for completely testing the application. Actually, the graphical interface is based on EditableGraph of NetGraph. The acquisition of a BooN is achieved by catching the events of EditableGraph. Effectively in this function I have also remarked that the edition is frozen by entering Return several times without pointing to a node. Hence, I cannot correct it by my own unless by fully defining a new version for acquiring network graphically, which is planned for 2025. However, I have already send a message to the developer of Netgraph concerning this issue.

* Sometimes if I try to edit an update function but don't actually change anything, the app will report a syntax error. One time, it even started repeating this error dialog after I closed it and had to be killed.

For the modification of the formula using the formula view you first need to shift from logical to Python because only the python syntax is parsed. I have now added this tip in the syntax error message to guide the user.

* Is there some sort of loading indicator for computing the steady states and attractors when a complex model is analyzed? I have not encountered one.

The stable state computation is very fast and based on a Z3 SAT Solver and I do not know if there is such indicator in the solver. For the attractor computation, this can be added and it will be an issue.

[Mussel2010] Müssel, Christoph, Martin Hopfensitz, and Hans A. Kestler. "BoolNet—an R package for generation, reconstruction and analysis of Boolean networks." *Bioinformatics* 26.10 (2010): 1378-1380.

[Klärner2017] Klärner, Hannes, Adam Streck, and Heike Siebert. "PyBoolNet: a python package for the generation, analysis and visualization of boolean networks." *Bioinformatics* 33.5 (2017): 770-772.

[Benes2022] Beneš, Nikola, et al. "AEON. py: Python library for attractor analysis in asynchronous Boolean networks." *Bioinformatics* 38.21 (2022): 4978-4980.

Reviewer: 3

Comments to the Author

1. General assessment of the work:

in this work, Delaplace et al. report the development of a Boolean Network analysis package called BooN. Overall the package seems to be an interesting proposition for the use of Boolean network models in biology (and other fields).

2. Specific comments for revision (mandatory):

- Although the software looks useful, I do not see what the software does that goes beyond what is already in the field. Is the novelty the ability to do the controllability analysis? Or is it the visualization?

- What is it that BooN does that others do not?

Mainly, BooN the novelty is the integration of the controllability analysis in a platform dedicated to Boolean network analysis. Moreover we can perform this analysis using the graphical interface. The other features are also in the other software related to Boolean network analysis. Moreover, as far as I know, there is no such development in python including a graphical interface for Boolean network manipulation. Therefore, BooN offer thus this development for the scientific community since the sources are provided in the distribution.

- The figure showing the number of publications per modeling framework further highlights the need to show why BooN is an improvement over all of these.

- The author mentions that Boolean evaluation can be synchronous or asynchronous. Can BooN do both?

Yes it can do both. I have added a sentence to insist on this fact.

- What does the controllability analysis enable in a bioinformatics context?

The controllability analysis is central for discovering the molecular causes of the diseases and predict drug targets. Mainly the controllability identifies the variable that must be frozen to True or False in order to reach an expected fate. The expected fate usually depicts a biomarker profile. Therefore the controllability analysis could answer to the following question: Given a Boolean network which genes must be frozen to cure a disease identified by a healthy biomark profiles.

- Figure 4 would be better as a workflow.

There is no real workflow but rather several services. I am sorry bu I do not understand by workflow here.

- How does one use BooN? A tutorial for use would be useful either with the paper and/or on the GitHub site.

A tutorial is available in the GitHub site ([Example/example.py](#)) showing the main features of Boon. Moreover, in README I also detail an interactive application of Boon using the GUI on Breast cancer study.

3. Specific comments for revision (optional):

- Figure 1 is uninformative.

I'm sincerely sorry I don't understand your comment. To what extent is the figure showing the number of publications in Pubmed with associated queries not informative?

- The conclusion reads more like an introduction, maybe start with that ?

Yes you are right it could. However, in conclusion we have also included a comparison with the current software, and we think it would be better to achieve this comparison after explaining Boon.

- The paper switches from high-level, to very specific maths, to high level in a way that makes it hard to follow. Given the broad readership of the journal, the authors may consider a supplement for the detailed info.

The controllability analysis is detailed in [Biane2109] I have added a sentence to invite the reader to refer to this article for more details.

- Visual Boon seems to be a really interesting tool. Could you tell us more about the tool and what the visualization does?

In the article, I have incorporated additional details regarding the graphical interfaces.

[Biane2109] Célia Biane, Franck Delaplace - Causal reasoning on Boolean control networks based on abduction: theory and application to Cancer drug discovery. IEEE/ACM Trans Comput Biol Bioinform. 2018 Dec 21.